Simulation Arena

# Deal Approval & Margin Balancing

**Approve 18% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $4.30M of expected profit.**

Seed 42 | Run run_825e3ed2d333 | 170,000 scenarios tested | Monte Carlo simulation with an empirica… | 21 Sep 2026, 20:10

Same seed, same numbers - this run can be reproduced exactly.

## How this works

For each discount level we ran thousands of 'what-if' worlds. In each one the buyer's odds of saying yes wobble a little, and your delivery cost wobbles a little. We then ask the only question that matters: if you offered this discount again and again, how much profit would you bank on average? That is the expected profit - the chance of winning multiplied by the profit you keep. Because those worlds are a sample, we also work out how much the answer itself would wobble if we ran it again. Any discount whose expected profit is within twice that wobble of the best - or within 1% of the best profit, whichever is wider - is treated as giving the same answer. That is the range where you have genuine room to negotiate.

## The numbers that matter

BEST DISCOUNT TO OFFER
**18%**
This is where the extra wins are still worth more than the margin given away.
vs rep asked for 25%

EXPECTED PROFIT AT THAT DISCOUNT
**$18.87M**
Average profit if you made this call many times over.

DISCOUNT RANGE YOU CAN APPROVE WITHOUT LOSING MONEY
**18%**
Only one discount stands out - the model cannot tell any neighbouring discount apart from it. The model's own margin of error here is about $429k, so treat this as the best single number - the next discount along already costs real money.
vs rep asked for 25% - outside the range

WHERE IT STARTS TO GENUINELY COST YOU
**20%**
At 20% the expected profit falls $396k below the best. That gap is wider than the model's own margin of error, so the loss is real.
vs costs about $396k

EXPECTED PROFIT IF YOU CAVE IN
**$14.57M**
What 25% actually earns you.
vs -$4.30M vs the optimum

PROFIT GIVEN AWAY
**$4.30M**
Buying +15 points of win probability costs you this much.

ODDS OF WINNING AT THE BEST DISCOUNT
**77.5%**
The chance the deal lands.

WORST CASE (1-IN-20)
**$0**
In an unlucky run of 20 deals, this is roughly the profit you would see.

## How we got there

### 1. The two forces fighting each other

Every point of discount does two things: it makes the buyer a bit more likely to say yes, and it takes $1.12M straight off your profit. Early on the first effect wins. Later, the second one does.

Evidence: Cost to deliver $67.91M on $111.58M of list value

### 2. What the buyer's behaviour looks like

They start around 35.0% likely to buy at full price and never become certain - the curve flattens near 95.0%. Around 15% discount is where they are most persuadable.

Evidence: Curve source: 450 of your past deals

**3. The sweet spot**

18% discount gives the highest expected profit at $18.87M, with a 77.5% chance of winning.

Evidence: peak of 17 levels tested

**4. The range where the answer is the same**

18% is the only discount that stands out; the model cannot tell any neighbouring discount apart from it. The next level along already costs $396k.

Evidence: margin of error $429k; safe range 18%

**5. What the requested discount really costs**

Moving from 18% to 25% buys +15 points of win probability but costs $4.30M of expected profit - more than the model's margin of error, so the loss is real.

Evidence: $14.57M vs $18.87M

## Charts

### Where profit peaks

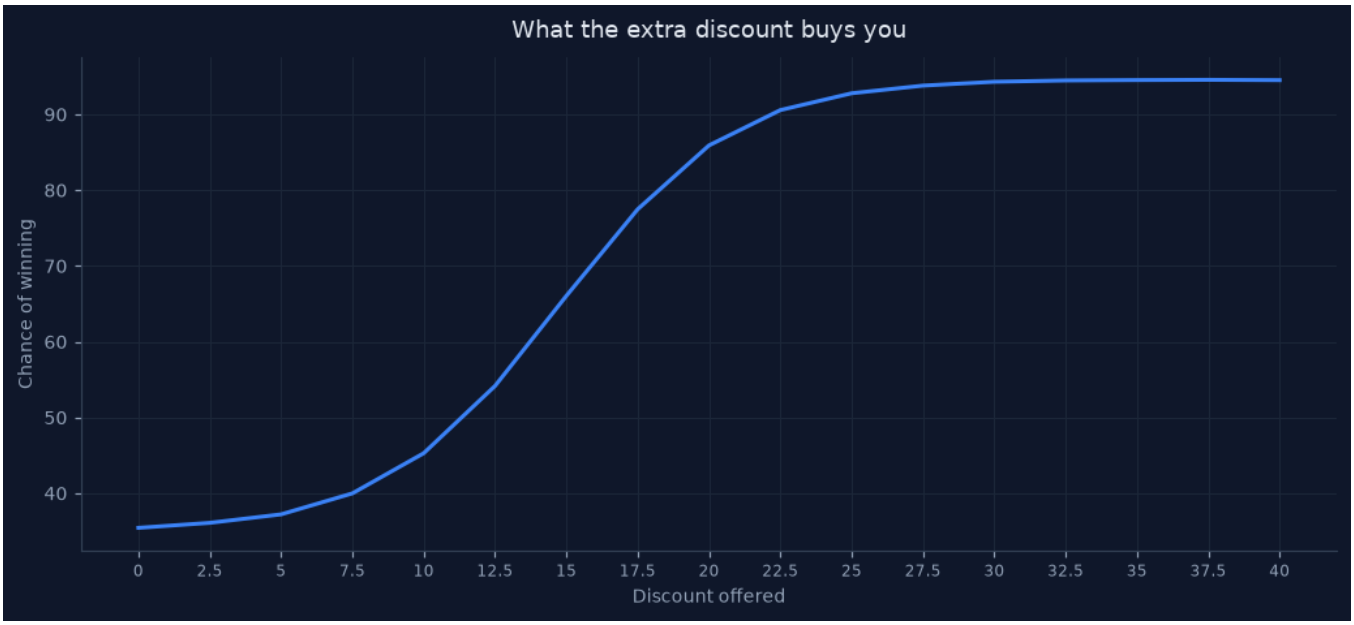
Everything from 18% gives the same answer within the model's margin of error; the peak is 18%




*What this shows: Grey is what you keep per won deal - it only falls. Green is what you actually expect to earn - it rises, peaks, then falls. The amber dashed line is the best profit minus the model's own margin of error: every discount whose green line stays above it is the same answer, which is the 18% range. Past that you are genuinely paying for the win.*

### What the extra discount buys you

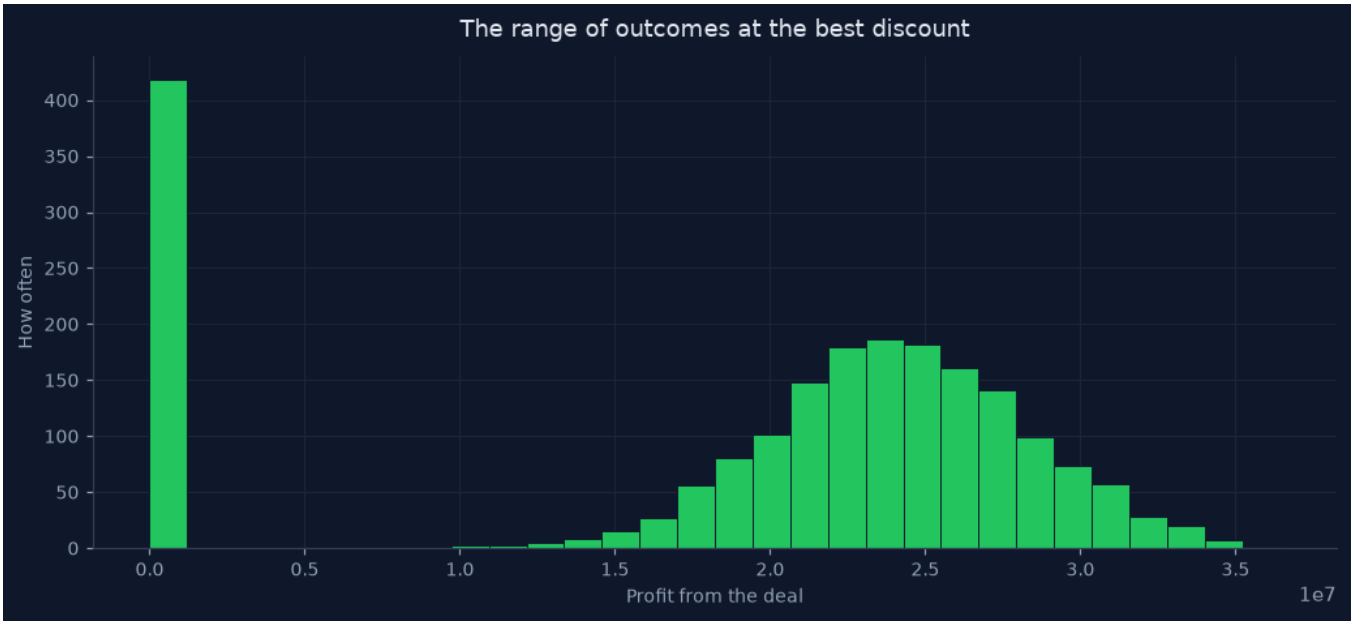
Curve learned from 450 of your past deals




*What this shows: Past the knee of this curve, extra discount buys almost no extra certainty - but keeps costing margin. That is the trap.*

### The range of outcomes at the best discount

10,000 simulated worlds at 18% discount




*What this shows: The cluster at zero is the deals you lose. The spread on the right is the profit you make when you win.*

## Tables

### Every discount level tested

Expected profit = chance of winning x profit kept. Highlighted row is the winner.

| Discount | Price after discount | Profit if won | Win chance | Expected profit | Worst case | Breaches floor? |
|---|---|---|---|---|---|---|
| 0.0% | $111,585,000 | $43,671,451 | 35.4% | $15,646,106 | $0 | 0% chance |
| 2.5% | $108,795,375 | $40,927,206 | 36.1% | $14,631,398 | $0 | 0% chance |
| 5.0% | $106,005,750 | $38,096,443 | 37.2% | $14,300,515 | $0 | 0% chance |
| 7.5% | $103,216,125 | $35,284,778 | 40.0% | $14,221,904 | $0 | 0% chance |
| 10.0% | $100,426,500 | $32,503,199 | 45.3% | $14,617,138 | $0 | 0% chance |
| 12.5% | $97,636,875 | $29,851,295 | 54.2% | $16,094,262 | $0 | 0% chance |

| Discount | Price after discount | Profit if won | Win chance | Expected profit | Worst case | Breaches floor? |
|---|---|---|---|---|---|---|
| 15.0% | $94,847,250 | $26,963,322 | 66.1% | $17,832,984 | $0 | 0% chance |
| 17.5% | $92,057,625 | $24,252,399 | 77.5% | $18,865,622 | $0 | 0% chance |
| 20.0% | $89,268,000 | $21,410,297 | 85.9% | $18,469,167 | $0 | 0% chance |
| 22.5% | $86,478,375 | $18,588,255 | 90.6% | $16,905,092 | $0 | 2% chance |
| 25.0% | $83,688,750 | $15,775,567 | 92.8% | $14,568,704 | $0 | 8% chance |
| 27.5% | $80,899,125 | $12,959,049 | 93.8% | $12,162,348 | $0 | 21% chance |
| 30.0% | $78,109,500 | $10,107,675 | 94.3% | $9,528,412 | $0 | 43% chance |
| 32.5% | $75,319,875 | $7,392,684 | 94.5% | $6,947,155 | $0 | 66% chance |
| 35.0% | $72,530,250 | $4,558,265 | 94.5% | $4,330,672 | -$1,971,519 | 85% chance |
| 37.5% | $69,740,625 | $1,898,703 | 94.6% | $1,801,412 | -$4,584,397 | 95% chance |
| 40.0% | $66,951,000 | -$957,538 | 94.5% | -$910,607 | -$7,512,832 | 99% chance |

Row 8 maximises expected profit.

### The decision in one line

What changes if you approve the discount being asked for.

| Column 1 | Best (18%) | Asked (25%) | Difference |
|---|---|---|---|
| Discount | 17.5% | 25.0% | +7.5 pts |
| Win chance | 77.5% | 92.8% | +15.3 pts |
| Profit if won | $24,252,399 | $15,775,567 | -$8,476,833 |
| Expected profit | $18,865,622 | $14,568,704 | -$4,296,918 |
| Range where the answer is the same | 18% | outside it | margin of error $429k |

### The range where the answer is the same

Every discount we tested, how far it falls short of the best, and whether that shortfall is real money or just the model's own margin of error.

| Discount | Expected profit | Short of the best | Inside the range? | What it means |
|---|---|---|---|---|
| 0.0% | $15,646,106 | $3,219,516 | Outside | costs you $3.22M |
| 2.5% | $14,631,398 | $4,234,224 | Outside | costs you $4.23M |
| 5.0% | $14,300,515 | $4,565,107 | Outside | costs you $4.57M |
| 7.5% | $14,221,904 | $4,643,719 | Outside | costs you $4.64M |
| 10.0% | $14,617,138 | $4,248,484 | Outside | costs you $4.25M |
| 12.5% | $16,094,262 | $2,771,360 | Outside | costs you $2.77M |
| 15.0% | $17,832,984 | $1,032,638 | Outside | costs you $1.03M |
| 17.5% | $18,865,622 | $0 | Inside | safe to approve |
| 20.0% | $18,469,167 | $396,455 | Outside | costs you $396k |
| 22.5% | $16,905,092 | $1,960,530 | Outside | costs you $1.96M |
| 25.0% | $14,568,704 | $4,296,918 | Outside | costs you $4.30M |
| 27.5% | $12,162,348 | $6,703,274 | Outside | costs you $6.70M |
| 30.0% | $9,528,412 | $9,337,210 | Outside | costs you $9.34M |
| 32.5% | $6,947,155 | $11,918,467 | Outside | costs you $11.92M |
| 35.0% | $4,330,672 | $14,534,951 | Outside | costs you $14.53M |
| 37.5% | $1,801,412 | $17,064,210 | Outside | costs you $17.06M |
| 40.0% | -$910,607 | $19,776,229 | Outside | costs you $19.78M |

Highlighted rows are the range where the answer is the same (18%) - within about $429k of the best.

### Recommended discount per deal (top 25)

The same logic applied to each deal individually.

| Deal | Customer | Value | Margin | Best discount | Expected profit |
|---|---|---|---|---|---|
| D-0413 | Tyrell Holdings | $1,922,000 | 45.1% | 20% | $414,250 |
| D-0251 | Globex Energy | $1,663,000 | 42.0% | 18% | $316,043 |
| D-0084 | Basalt Motors | $1,395,000 | 39.2% | 18% | $234,813 |
| D-0312 | Verity Group | $1,383,000 | 42.7% | 18% | $270,341 |
| D-0214 | Vehement Energy | $1,309,000 | 30.4% | 0% | $141,078 |
| D-0091 | Wayne Retail | $1,104,000 | 38.4% | 18% | $178,980 |
| D-0103 | Hooli Systems | $1,084,000 | 31.5% | 0% | $121,056 |
| D-0307 | Kestrel Bank | $1,025,000 | 35.9% | 18% | $146,295 |
| D-0048 | Duff Group | $995,000 | 30.3% | 0% | $106,884 |
| D-0053 | Ironwood Systems | $985,000 | 38.6% | 18% | $161,216 |
| D-0419 | Vehement Media | $984,000 | 34.4% | 18% | $128,994 |
| D-0203 | Vandelay Media | $922,000 | 35.8% | 18% | $130,879 |
| D-0355 | Initech Logistics | $858,000 | 34.7% | 18% | $114,473 |
| D-0269 | Gringotts Group | $856,000 | 28.3% | 0% | $85,883 |
| D-0302 | Massive Motors | $827,000 | 33.4% | 18% | $101,998 |
| D-0068 | Tessera Group | $820,000 | 38.5% | 18% | $133,574 |
| D-0428 | Prestige Partners | $811,000 | 45.9% | 20% | $180,367 |
| D-0195 | Vandelay Chemicals | $798,000 | 37.3% | 18% | $122,562 |
| D-0028 | Initech Logistics | $756,000 | 48.2% | 20% | $183,065 |
| D-0364 | Verity Systems | $743,000 | 39.6% | 18% | $127,371 |
| D-0430 | Prestige Holdings | $738,000 | 33.8% | 18% | $93,311 |
| D-0323 | Gringotts Energy | $721,000 | 32.2% | 0% | $82,307 |
| D-0004 | Hooli Freight | $697,000 | 35.4% | 18% | $96,777 |
| D-0444 | Cyberdyne Freight | $693,000 | 39.1% | 18% | $116,112 |
| D-0317 | Cobalt Labs | $677,000 | 39.7% | 18% | $116,582 |

## What we recommend

**Approve 18% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $4.30M of expected profit.**

**Risk Low** 25 out of 100 | Confidence: High

### Why

• Only 18% stands out from the rest - the model cannot tell any neighbouring discount apart from it.

• Expected profit peaks at 18% ($18.87M).

• The requested 25% delivers $14.57M - $4.30M less than the peak, which is more than the model's margin of error.

• Win probability only improves +15 points for that giveaway.

### What to do

**1.** Approve 18% as the standard concession - it is the only discount the model can tell apart from the rest.

**2.** Above 18% (the top of the safe range), require VP sign-off: the 25% being asked for costs about $573k per point of discount.

**3.** If this quarter must not misfire, 0% is the safer call: it maximises the bad case rather than the average.

**The trade-off:** You are trading $4.30M of expected profit for +15 points of win probability.

## What we assumed and what to watch

### Assumptions

• List value $111,585,000, cost $67,906,095 (39.1% standard margin).

• Consumption effects: none - your file has no tier accelerator, burst or rollover columns, so this is a straight price-versus-margin question.

• Win probability curve: from 35.0% at full price toward a ceiling of 95.0%, halfway at 14.8% discount (source: 450 of your past deals).

• Win probability uncertainty: plus or minus 5 points.

• Cost volatility: plus or minus 6% of cost.

• Margin floor: 12.0%.

• Two discounts are treated as giving the same answer when the gap between their expected profits is smaller than twice the model's own margin of error at both discounts added together (about $429k at the best one), or smaller than 1% of the best profit - whichever is larger.

• The range where the answer is the same is 18%, taken from the 17 discount levels we tested.

• 10,000 simulations per discount level; seed 42 (re-running gives identical numbers).

• Deal / quote ID -> your 'deal_id' column

• Customer name -> your 'customer' column

• Deal value (list price) -> your 'deal_value' column

• Cost to deliver -> your 'cost' column

• Standard margin (%) -> your 'margin_pct' column

• Discount given (%) -> your 'discount_pct' column

• Won? (1/0) -> your 'won' column

• Hit a consumption tier accelerator -> not in your file; not used

• Unmetered burst used -> not in your file; not used

• Rollover window -> not in your file; not used

## Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

| Time | Stage | What happened |
|---|---|---|
| 0.00s | prepare | Discount given (%) -> your 'discount_pct' column |
| 0.00s | prepare | Won? (1/0) -> your 'won' column |
| 0.00s | prepare | Hit a consumption tier accelerator -> not in your file; not used |
| 0.00s | prepare | Unmetered burst used -> not in your file; not used |
| 0.00s | prepare | Rollover window -> not in your file; not used |
| 0.01s | prepare | Read 450 records into the model. |
| 0.01s | simulate | Running the simulation… |
| 0.01s | read | Reading the deal you gave us... |
| 0.01s | read | Found 450 deal(s) worth $111.58M at an average 39.1% margin. |
| 0.01s | prepare | Cost to deliver is about $67.91M. |
| 0.01s | prepare | Working out how much a discount really moves the buyer... |
| 0.03s | prepare | Learned the buyer's behaviour from 450 past deals (they flip hardest around 14.8% discount). |
| 0.03s | simulate | Testing 17 discount levels, 10,000 outcomes each (170,000 scenarios in total)... |

| Time | Stage | What happened |
|---|---|---|
| 0.03s | simulate | Simulated 11% of the discount levels... |
| 0.03s | simulate | Simulated 23% of the discount levels... |
| 0.03s | simulate | Simulated 35% of the discount levels... |
| 0.04s | simulate | Simulated 47% of the discount levels... |
| 0.04s | simulate | Simulated 58% of the discount levels... |
| 0.04s | simulate | Simulated 70% of the discount levels... |
| 0.04s | simulate | Simulated 82% of the discount levels... |
| 0.04s | simulate | Simulated 94% of the discount levels... |
| 0.04s | simulate | Simulated 100% of the discount levels... |
| 0.04s | summarize | All discount levels tested. Looking for the sweet spot... |
| 0.04s | summarize | Checked how much the answer would wobble if we ran it again - the range where the answer is the same is 18%. |
| 0.04s | summarize | Done. Best discount is 18% ($18.87M expected profit); the range where the answer is the same is 18%. |